Optimal Exit Time for Liquidity Providers in Automated Market Makers

Philippe Bergault[1], Sébastien Bieber[2], Leandro Sánchez-Betancourt[3]

London – Paris Bachelier Workshop, November 2025

[1,2]Université Paris-Dauphine PSL,[3]Oxford-Man Institute of Quantitative Finance.

Introduction

Automated Market Makers (AMMs)

AMM

Protocol that lets users trade and provide liquidity directly through smart contracts, with prices determined by mathematical algorithms.

Automated Market Makers (AMMs)

AMM

Protocol that lets users trade and provide liquidity directly through smart contracts, with prices determined by mathematical algorithms.

Key Features:

Automated Market Makers (AMMs)

AMM

Protocol that lets users trade and provide liquidity directly through smart contracts, with prices determined by mathematical algorithms.

Key Features:

- Agents: Liquidity Providers (LP) deposit a pair of assets; Liquidity Takers (LTs) swap one asset against another.
- Price discovery: Uses pre-defined and public algorithms instead of order books or human market makers.
- Transparency: Liquidity in the pool is always publicly visible.

Note

CFMMs are the most popular type of AMMs

Constant Function Market Makers (CFMM)

CFMM

CFMMs are characterized by a trading function $f : \mathbb{R}_+^* \times \mathbb{R}_+^* \to \mathbb{R}$.

- If q^0 and q^1 are reserves of assets 0 and 1, a swap of Δq^1 coins of asset 1 for Δq^0 coins of asset 0 satisfies:

$$f(q^0 - \Delta q^0, q^1 + \Delta q^1) = f(q^0, q^1)$$

$\rightarrow$ The function $f(q^0, q^1)$ is preserved in a trade (ignoring transaction costs).

Constant Function Market Makers (CFMM)

CFMM

CFMMs are characterized by a trading function $f : \mathbb{R}_+^* \times \mathbb{R}_+^* \to \mathbb{R}$.

- If q^0 and q^1 are reserves of assets 0 and 1, a swap of Δq^1 coins of asset 1 for Δq^0 coins of asset 0 satisfies:

$$f(q^0 - \Delta q^0, q^1 + \Delta q^1) = f(q^0, q^1)$$

$\to$ The function $f(q^0, q^1)$ is preserved in a trade (ignoring transaction costs).

Constant Product Market Makers (CPMM)

The most common CFMM is the Constant Product Market Maker, where

$$f(q^0, q^1) = q^0 q^1$$

Model & Problem Formulation

Model Setup

Liquidity Pool Setup

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Some notations

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Some notations

- X_t, Y_t: quantities of assets X and Y in the pool at time $t \leq T$,

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Some notations

- X_t, Y_t: quantities of assets X and Y in the pool at time $t \leq T$,
- Z_t: local price of Y in terms of X at time $t \leq T$,

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Some notations

- X_t, Y_t: quantities of assets X and Y in the pool at time $t \leq T$,
- Z_t: local price of Y in terms of X at time $t \leq T$,
- S_t: external mid-price of asset Y in terms of asset X at time $t \leq T$,

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Some notations

- X_t, Y_t: quantities of assets X and Y in the pool at time $t \leq T$,
- Z_t: local price of Y in terms of X at time $t \leq T$,
- S_t: external mid-price of asset Y in terms of asset X at time $t \leq T$,
- N_t^a, N_t^b: number of liquidity taking buys, sells at time $t \leq T$,

Model Setup

Liquidity Pool Setup

- A LP deposits a pair of assets X (e.g., USDC) and Y (e.g.; ETH) for a time horizon $T > 0$ into a liquidity pool with the CPMM function.
- We assume the existence of an external limit order book venue for trading X and Y, where price formation occurs.
- LTs arrive according to the counting processes N^a and N^b.

Some notations

- X_t, Y_t: quantities of assets X and Y in the pool at time $t \leq T$,
- Z_t: local price of Y in terms of X at time $t \leq T$,
- S_t: external mid-price of asset Y in terms of asset X at time $t \leq T$,
- N_t^a, N_t^b: number of liquidity taking buys, sells at time $t \leq T$,
- ξ: trade size.

Dynamics of Reserves and Price

Dynamics of reserves X, Y, and local price Z

Dynamics of Reserves and Price

Dynamics of reserves X, Y, and local price Z

$$\mathrm{d}Y_t = \xi\,\mathrm{d}N_t^b - \xi\,\mathrm{d}N_t^a,$$
$$\mathrm{d}X_t = [\varphi_c(Y_{t\text{-}} + \xi) - \varphi_c(Y_{t\text{-}})]\,\mathrm{d}N_t^b + [\varphi_c(Y_{t\text{-}} - \xi) - \varphi_c(Y_{t\text{-}})]\,\mathrm{d}N_t^a,$$
$$\mathrm{d}Z_t = [-\varphi_c'(Y_{t\text{-}} + \xi) + \varphi_c'(Y_{t\text{-}})]\,\mathrm{d}N_t^b + [-\varphi_c'(Y_{t\text{-}} - \xi) + \varphi_c'(Y_{t\text{-}})]\,\mathrm{d}N_t^a.$$

Cartea et al. (2024)

Dynamics of Reserves and Price

Dynamics of reserves X, Y, and local price Z

$$\mathrm{d}Y_t = \xi\,\mathrm{d}N_t^b - \xi\,\mathrm{d}N_t^a,$$
$$\mathrm{d}X_t = [\varphi_c(Y_{t\text{-}} + \xi) - \varphi_c(Y_{t\text{-}})]\,\mathrm{d}N_t^b + [\varphi_c(Y_{t\text{-}} - \xi) - \varphi_c(Y_{t\text{-}})]\,\mathrm{d}N_t^a,$$
$$\mathrm{d}Z_t = [-\varphi_c'(Y_{t\text{-}} + \xi) + \varphi_c'(Y_{t\text{-}})]\,\mathrm{d}N_t^b + [-\varphi_c'(Y_{t\text{-}} - \xi) + \varphi_c'(Y_{t\text{-}})]\,\mathrm{d}N_t^a.$$

Cartea et al. (2024)

Note

We define $\varphi_c(y) = \frac{c}{y}$ as the level function with constant depth c throughout the trading horizon. The depth c is determined at deposit inception and depends on the initial reserves ($c = X_0 Y_0$).

Dynamics of Reserves and Price

Dynamics of reserves X, Y, and local price Z

$$\mathrm{d}Y_t = \xi\,\mathrm{d}N_t^b - \xi\,\mathrm{d}N_t^a,$$
$$\mathrm{d}X_t = [\varphi_c(Y_{t\text{-}} + \xi) - \varphi_c(Y_{t\text{-}})]\,\mathrm{d}N_t^b + [\varphi_c(Y_{t\text{-}} - \xi) - \varphi_c(Y_{t\text{-}})]\,\mathrm{d}N_t^a,$$
$$\mathrm{d}Z_t = [-\varphi_c'(Y_{t\text{-}} + \xi) + \varphi_c'(Y_{t\text{-}})]\,\mathrm{d}N_t^b + [-\varphi_c'(Y_{t\text{-}} - \xi) + \varphi_c'(Y_{t\text{-}})]\,\mathrm{d}N_t^a.$$

Cartea et al. (2024)

Note

We define $\varphi_c(y) = \frac{c}{y}$ as the level function with constant depth c throughout the trading horizon. The depth c is determined at deposit inception and depends on the initial reserves ($c = X_0 Y_0$).

Dynamics of External Mid-Price S

Dynamics of Reserves and Price

Dynamics of reserves X, Y, and local price Z

$$\mathrm{d}Y_t = \xi\,\mathrm{d}N_t^b - \xi\,\mathrm{d}N_t^a,$$
$$\mathrm{d}X_t = [\varphi_c(Y_{t^-} + \xi) - \varphi_c(Y_{t^-})]\,\mathrm{d}N_t^b + [\varphi_c(Y_{t^-} - \xi) - \varphi_c(Y_{t^-})]\,\mathrm{d}N_t^a,$$
$$\mathrm{d}Z_t = [-\varphi_c'(Y_{t^-} + \xi) + \varphi_c'(Y_{t^-})]\,\mathrm{d}N_t^b + [-\varphi_c'(Y_{t^-} - \xi) + \varphi_c'(Y_{t^-})]\,\mathrm{d}N_t^a.$$

Cartea et al. (2024)

Note

We define $\varphi_c(y) = \frac{c}{y}$ as the level function with constant depth c throughout the trading horizon. The depth c is determined at deposit inception and depends on the initial reserves ($c = X_0 Y_0$).

Dynamics of External Mid-Price S

$$S_t = S_0 + \sigma W_t, \quad \text{where} \quad S_0 \in \mathbb{R}^+.$$

Designing Stochastic Intensities

We suppose two types of LTs interacting with the liquidity pool:

Designing Stochastic Intensities

We suppose two types of LTs interacting with the liquidity pool:

- **Arbitrageurs**: Look for profit from the price difference between the external venue and the AMM, thus the difference between S and Z.

Designing Stochastic Intensities

We suppose two types of LTs interacting with the liquidity pool:

- **Arbitrageurs**: Look for profit from the price difference between the external venue and the AMM, thus the difference between S and Z.
- **Noise Traders**: They trade without regard to the price difference, usually for other purposes.

Designing Stochastic Intensities

We suppose two types of LTs interacting with the liquidity pool:

- **Arbitrageurs**: Look for profit from the price difference between the external venue and the AMM, thus the difference between S and Z.
- **Noise Traders**: They trade without regard to the price difference, usually for other purposes.

Stochastic Intensities

Designing Stochastic Intensities

We suppose two types of LTs interacting with the liquidity pool:

- **Arbitrageurs**: Look for profit from the price difference between the external venue and the AMM, thus the difference between S and Z.
- **Noise Traders**: They trade without regard to the price difference, usually for other purposes.

Stochastic Intensities

$$\lambda_t^a = \bar{\lambda}^a(Z_{t^-}, S_t) = \max\{a_0,\ a_1 + a_2 \cdot (S_t - Z_{t^-})\},$$
$$\lambda_t^b = \bar{\lambda}^b(Z_{t^-}, S_t) = \max\{a_0,\ a_1 + a_2 \cdot (Z_{t^-} - S_t)\}.$$

- $a_0 > 0$: minimum base intensity.
- $a_1, a_2 \geq 0$: coefficients that modulate intensity based on price differences.

Liquidity Provider Problem

Fees

Liquidity Provider Problem

Fees

Payments from LTs to the LP for executing trades as compensation for price risk and liquidity provision.

$$R_t = \int_0^t \mathfrak{r}(Y_{s^-}) \left(\mathrm{d}N_s^b + \mathrm{d}N_s^a \right) \quad \text{with} \quad \mathfrak{r} : \mathbb{R}_+ \to \mathbb{R}_+.$$

Impermanent Loss (IL)

Liquidity Provider Problem

Fees

Payments from LTs to the LP for executing trades as compensation for price risk and liquidity provision.

$$R_t = \int_0^t \mathfrak{r}(Y_{s^-}) \left(\mathrm{d}N_s^b + \mathrm{d}N_s^a\right) \quad \text{with} \quad \mathfrak{r} : \mathbb{R}_+ \to \mathbb{R}_+.$$

Impermanent Loss (IL)

Reduction in value a LP experiences when the prices of the assets in the pool change, compared to simply holding those assets outside the pool.

$$\mathrm{IL}_t := -\left[X_t + Y_t S_t - (X_0 + Y_0 S_t)\right] = -\Big[\underbrace{X_t - X_0}_{P_t^X} + S_t \underbrace{(Y_t - Y_0)}_{P_t^Y}\Big].$$

Liquidity Provider Problem

Fees

Payments from LTs to the LP for executing trades as compensation for price risk and liquidity provision.

$$R_t = \int_0^t \mathfrak{r}(Y_{s^-}) \left(\mathrm{d}N_s^b + \mathrm{d}N_s^a\right) \quad \text{with} \quad \mathfrak{r} : \mathbb{R}_+ \to \mathbb{R}_+.$$

Impermanent Loss (IL)

Reduction in value a LP experiences when the prices of the assets in the pool change, compared to simply holding those assets outside the pool.

$$\mathrm{IL}_t := -\left[X_t + Y_t S_t - (X_0 + Y_0 S_t)\right] = -\Big[\underbrace{X_t - X_0}_{P_t^X} + S_t \underbrace{(Y_t - Y_0)}_{P_t^Y}\Big].$$

Note

We consider the case of a LP who owns the entire liquidity available in the AMM and that $\mathfrak{r}$ has linear growth.

Liquidity Provider Problem

LP's optimal problem

The LP wants to maximise her fee revenues while minimizing her IL.

Let $\mathcal{T}_{t,T}$ be set of stopping times taking values in $[t, T]$, and $\mathcal{T} := \mathcal{T}_{0,T}$.

The LP optimal problem is:

$$\sup_{\tau \in \mathcal{T}} \mathbb{E}\Big[\underbrace{P_\tau^X + S_\tau P_\tau^Y}_{-\mathrm{IL}_\tau} + R_\tau\Big].$$

Value function and Dynamic Programming Principle

We define the value function v for this optimal stopping time problem as the following:

$v : [0, T] \times Q \times \mathbb{R} \longrightarrow \mathbb{R}$,

$$(t, y, S) \longmapsto \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}\Bigg[\int_t^{\tau} \Bigg\{ \overbrace{\left(\varphi_c(Y_{s^-}^{t,y} + \xi) - \varphi_c(Y_{s^-}^{t,y}) + \xi S_s^{t,S} + \mathfrak{r}(Y_{s^-}^{t,y})\right)}^{:=\beta^b(Y_{s^-}^{t,y}, S_s^{t,S})} \lambda_s^b$$

$$+ \underbrace{\left(\varphi_c(Y_{s^-}^{t,y} - \xi) - \varphi_c(Y_{s^-}^{t,y}) - \xi S_s^{t,S} + \mathfrak{r}(Y_{s^-}^{t,y})\right)}_{:=\beta^a(Y_{s^-}^{t,y}, S_s^{t,S})} \lambda_s^a \Bigg\} \, \mathrm{d}s \Bigg].$$

Value function and Dynamic Programming Principle

We define the value function v for this optimal stopping time problem as the following:

$v : [0, T] \times Q \times \mathbb{R} \longrightarrow \mathbb{R}$,

$$(t, y, S) \longmapsto \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}\Bigg[\int_t^{\tau} \Bigg\{ \overbrace{\left(\varphi_c(Y_{s^-}^{t,y} + \xi) - \varphi_c(Y_{s^-}^{t,y}) + \xi S_s^{t,S} + \mathfrak{r}(Y_{s^-}^{t,y})\right)}^{:=\beta^b(Y_{s^-}^{t,y}, S_s^{t,S})} \lambda_s^b$$

$$+ \underbrace{\left(\varphi_c(Y_{s^-}^{t,y} - \xi) - \varphi_c(Y_{s^-}^{t,y}) - \xi S_s^{t,S} + \mathfrak{r}(Y_{s^-}^{t,y})\right)}_{:=\beta^a(Y_{s^-}^{t,y}, S_s^{t,S})} \lambda_s^a \Bigg\} \mathrm{d}s \Bigg].$$

Dynamic Programming Principle

Value function and Dynamic Programming Principle

We define the value function v for this optimal stopping time problem as the following:

$v : [0, T] \times Q \times \mathbb{R} \longrightarrow \mathbb{R}$,

$$(t, y, S) \longmapsto \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}\Bigg[\int_t^\tau \Bigg\{ \overbrace{\left(\varphi_c(Y_{s^-}^{t,y} + \xi) - \varphi_c(Y_{s^-}^{t,y}) + \xi S_s^{t,S} + \mathfrak{r}(Y_{s^-}^{t,y})\right)}^{:=\beta^b(Y_{s^-}^{t,y}, S_s^{t,S})} \lambda_s^b$$

$$+ \underbrace{\left(\varphi_c(Y_{s^-}^{t,y} - \xi) - \varphi_c(Y_{s^-}^{t,y}) - \xi S_s^{t,S} + \mathfrak{r}(Y_{s^-}^{t,y})\right)}_{:=\beta^a(Y_{s^-}^{t,y}, S_s^{t,S})} \lambda_s^a \Bigg\} \, \mathrm{d}s \Bigg].$$

Dynamic Programming Principle

Let $(t, y, S) \in [0, T) \times Q \times \mathbb{R}$, then for all $\theta \in \mathcal{T}_{t,T}$ we have

$$v(t, y, S) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}\Bigg[\int_t^{\tau \wedge \theta} \left[\beta^b(Y_{u^-}^{t,y}, S_u^{t,S}) \lambda_u^b + \beta^a(Y_{u^-}^{t,y}, S_u^{t,S}) \lambda_u^a \right] \mathrm{d}u$$

$$+ \mathbb{1}_{\{\tau \geq \theta\}} v(\theta, Y_\theta^{t,y}, S_\theta^{t,y}) \Bigg].$$

Hamilton–Jacobi–Bellman Quasi–Variational Equation

HJB QVI

Hamilton–Jacobi–Bellman Quasi–Variational Equation

HJB QVI

$$
\begin{aligned}
0 = \min \Big\{ & -\frac{\partial}{\partial t} v(t,y,S) - \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial S^2} v(t,y,S) \\
& - \bar{\lambda}^b(y,S)\big[\beta^b(y,S) + v(t,y+\xi,S) - v(t,y,S)\big] \\
& - \bar{\lambda}^a(y,S)\big[\beta^a(y,S) + v(t,y-\xi,S) - v(t,y,S)\big] , \\
& v(t,y,S)\Big\} \qquad \text{on } [0,T) \times Q \times \mathbb{R},
\end{aligned}
$$

with terminal condition $v(T,y,S) = 0$ for all $(y,S) \in Q \times \mathbb{R}$.

Hamilton–Jacobi–Bellman Quasi–Variational Equation

HJB QVI

$$
\begin{aligned}
0 = \min \Big\{ & -\frac{\partial}{\partial t} v(t,y,S) - \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial S^2} v(t,y,S) \\
& - \bar{\lambda}^b(y,S)\big[\beta^b(y,S) + v(t,y+\xi,S) - v(t,y,S)\big] \\
& - \bar{\lambda}^a(y,S)\big[\beta^a(y,S) + v(t,y-\xi,S) - v(t,y,S)\big], \\
& v(t,y,S)\Big\} \qquad \text{on } [0,T) \times Q \times \mathbb{R},
\end{aligned}
$$

with terminal condition $v(T,y,S) = 0$ for all $(y,S) \in Q \times \mathbb{R}$.

Note

The problem is composed of two state variables, which makes it tractable for numerical simulation.

Numerical Simulation

Iterative solving methods

Euler method

Iterative solving methods

Euler method

The HJB QVI combines diffusion and jump terms. At each time step δ, jump terms are treated explicitly, then diffusion is updated implicitly.

Iterative solving methods

Euler method

The HJB QVI combines diffusion and jump terms. At each time step δ, jump terms are treated explicitly, then diffusion is updated implicitly.

Recursive scheme for Longstaff–Schwartz method

Iterative solving methods

Euler method

The HJB QVI combines diffusion and jump terms. At each time step δ, jump terms are treated explicitly, then diffusion is updated implicitly.

Recursive scheme for Longstaff–Schwartz method

We can approximate the solution numerically by leveraging the DPP at each time step δ:

$$\hat{v}(t,y,S) = \max\left\{\mathbb{E}\left[\int_t^{t+\delta} \left\{\beta^b\left(Y_{s^-}^{t,y}, S_s^{t,S}\right)\lambda_s^b + \beta^a\left(Y_{s^-}^{t,y}, S_s^{t,S}\right)\lambda_s^a\right\} \mathrm{d}s \right.\right.$$
$$\left.\left. + \hat{v}\left(t+\delta, Y_{t+\delta}^{t,y}, S_{t+\delta}^{t,S}\right)\right], 0\right\},$$

with the terminal condition $\hat{v}(T,y,S) = 0$.

Comparison between Euler and Longstaff–Schwartz methods

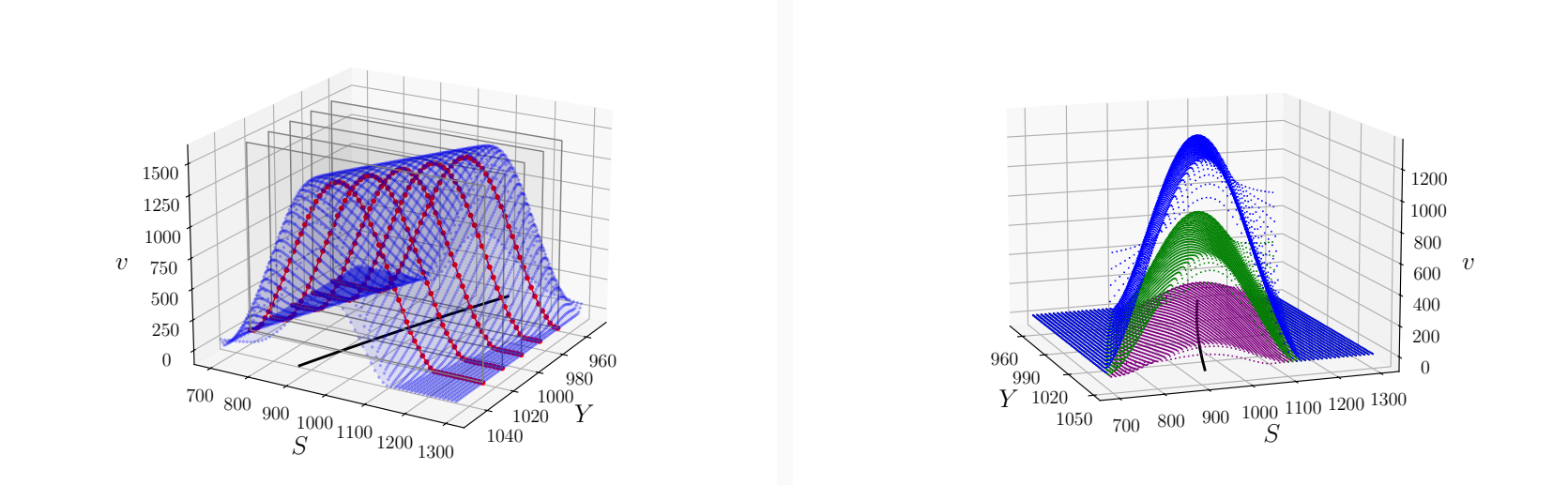

Figure 1: Left: value function v at $t = 0$ computed with the Euler method (blue surface) and Longstaff–Schwartz algorithm (red slices) with the black curve in the S-Y plane representing states where $S = Z$. Right: Euler method value function at $t = 0$ (blue), $t = 0.5$ (green), and $t = 0.9$ (purple).

Note

Fictitious parameters are used to ensure the stability of the Euler scheme.

Calibration on real market data

Market simulator

- Asset X: **USDC**, Asset Y: **ETH**
- Initial exchange rate: $S_0 = Z_0 = \mathbf{2820}\,\$$
- Initial reserve: $Y_0 = \mathbf{50{,}000}$ ETH, $X_0 = Y_0 Z_0\,\$$
- Volatility: $\sigma = \mathbf{0.0569}\, S_0\,\$ \cdot \text{day}^{-1/2}$
- Intensities: $a_0 = \mathbf{1}\ \text{day}^{-1}$, $a_1 = \mathbf{10}\ \text{day}^{-1}$, $a_2 = \mathbf{10}\ \$^{-1} \cdot \text{day}^{-1}$
- Trade size: $\xi = \mathbf{100}$
- Fees: $\mathfrak{r} = \mathbf{0.01} \times \xi \times S_0$
- Horizon: $T = \mathbf{1}$ day

Note

We use market data from Aqsha et al. (2025), calibrated with Binance and Uniswap V2 ETH-USDC data from January to April 2022.

Sample path for liquidity pool price and reserve

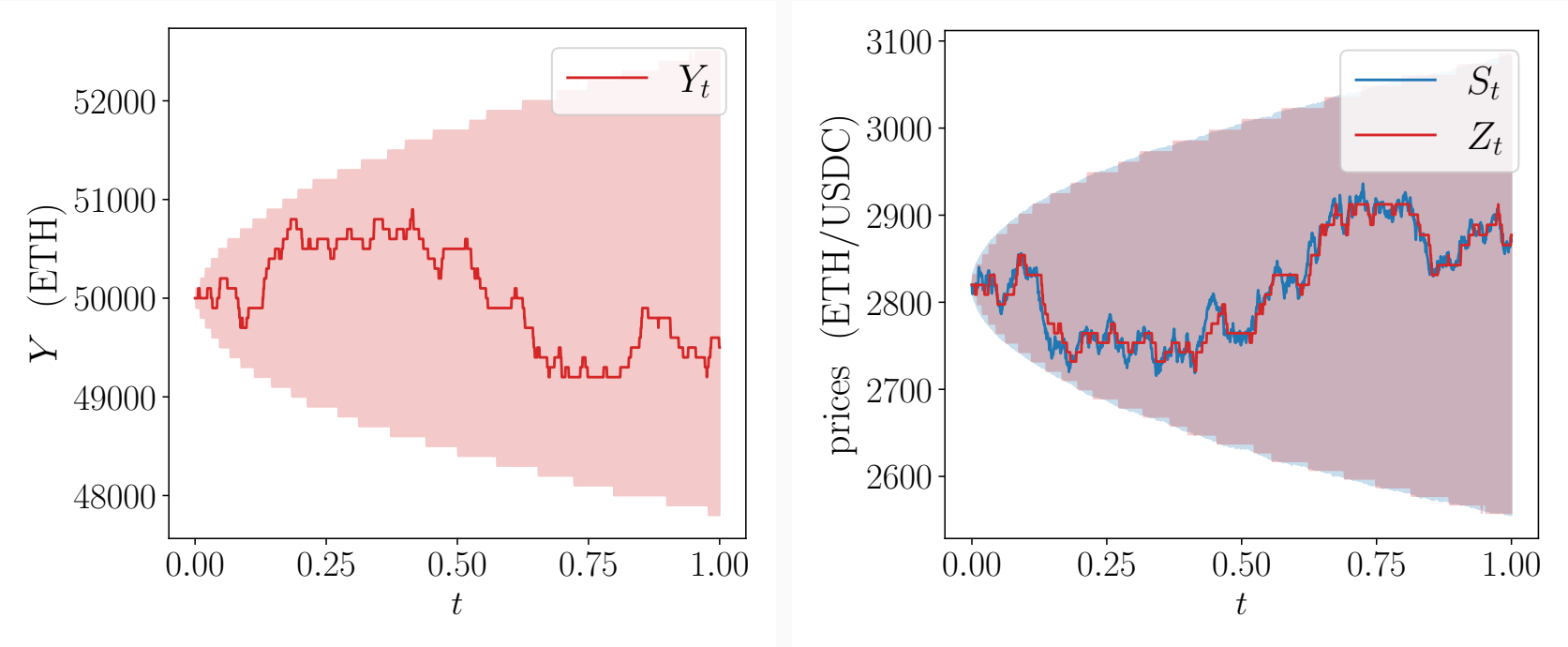

Figure 2: Sample paths for the inventory process Y_t (ETH) and the price processes S_t and Z_t. All processes are accompanied by the running quantiles (5% and 95%) across time.

Impact of volatility changes

σ_{new}	$\mathbb{E}[\tau]$	$\mathbb{E}[R_\tau]$	$\mathbb{E}[\text{IL}_\tau]$
$\sigma/5$	1.00 (0.00)	161,765 (22,320)	4,364 (6,419)
$\sigma/4$	1.00 (0.00)	168,343 (22,217)	6,912 (10,024)
$\sigma/3$	1.00 (0.00)	181,148 (21,806)	12,422 (17,830)
$\sigma/2$	1.00 (0.00)	210,406 (22,535)	28,177 (40,154)
σ	1.00 (0.01)	316,222 (26,930)	112,635 (157,844)
$2\,\sigma$	0.51 (0.27)	273,003 (144,005)	202,999 (372,645)
$3\,\sigma$	0.01 (0.01)	3,599 (9,363)	2,449 (13,042)
$4\,\sigma$	0.00 (0.00)	1,794 (3,984)	1,304 (7,411)
$5\,\sigma$	0.00 (0.00)	532 (1,198)	420 (1,401)

Table 1: Summary statistics for the expected (i) exit time,[1] (ii) fees collected, and (iii) impermanent loss, as σ varies. Mean values (with standard deviation in parenthesis) across 10,000 simulations.

[1] $\tau = T = 1$ if the LP does not exit the pool before T.

Impact of fees changes

Higher fees discourage LTs activity, so set $\tilde{a}_i = a_i\, e^{-\kappa(\tilde{\mathfrak{r}}-\mathfrak{r})}$, $i \in \{0, 1, 2\}$ to accurately evaluate the effect of fees.

$\mathfrak{r}_{\text{new}}$	$\mathbb{E}[\tau]$	$\mathbb{E}[R_\tau]$	$\mathbb{E}[\text{IL}_\tau]$
$\mathfrak{r}/5$	0.01 (0.01)	373 (750)	173 (1,308)
$\mathfrak{r}/4$	0.04 (0.05)	3,027 (5,099)	2,239 (11,377)
$\mathfrak{r}/3$	0.11 (0.12)	12,578 (14,578)	8,437 (35,736)
$\mathfrak{r}/2$	0.96 (0.09)	166,560 (20,956)	105,643 (141,534)
$\mathfrak{r}$	1.00 (0.01)	316,222 (26,930)	112,635 (157,844)
$2\,\mathfrak{r}$	1.00 (0.00)	530,160 (50,735)	113,269 (161,299)
$3\,\mathfrak{r}$	1.00 (0.00)	670,137 (71,973)	113,126 (161,712)
$4\,\mathfrak{r}$	1.00 (0.00)	755,901 (91,748)	112,825 (161,819)
$5\,\mathfrak{r}$	1.00 (0.00)	802,518 (109,834)	112,407 (161,782)

Table 2: Summary statistics for the expected (i) exit time, (ii) fees collected, and (iii) impermanent loss, as $\mathfrak{r}$ varies. Mean values (with standard deviation in parenthesis) across 10,000 simulations.

Impact of fees changes

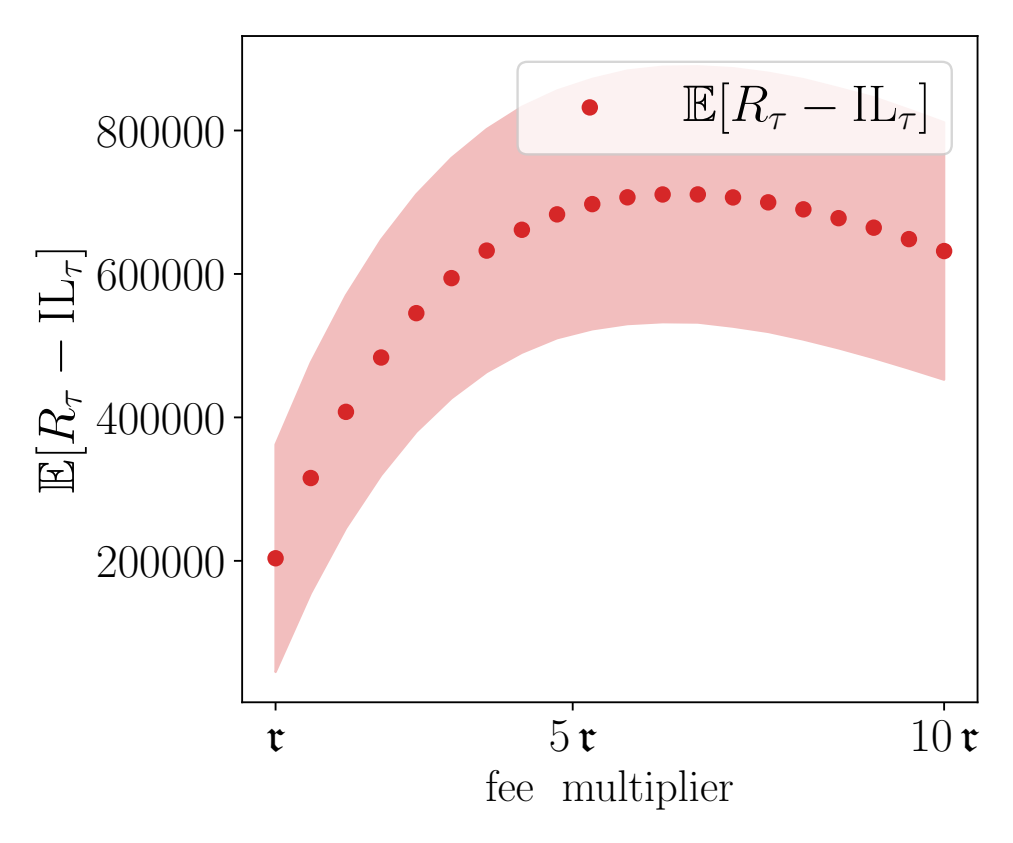

Figure 3: Relationship between $\mathfrak{r}$ and performance, measured through $\mathbb{E}[R_\tau - \mathrm{IL}_\tau]$. The coloured area shows one standard deviation around the mean.

Impact of intensity changes (a_1)

a_1^{new}	$\mathbb{E}[\tau]$	$\mathbb{E}[R_\tau]$	$\mathbb{E}[\text{IL}_\tau]$
$a_1/5$	1.00 (0.01)	296,407 (26,305)	112,619 (157,946)
$a_1/4$	1.00 (0.01)	297,518 (26,328)	112,633 (157,824)
$a_1/3$	1.00 (0.01)	299,641 (26,392)	112,652 (158,199)
$a_1/2$	1.00 (0.01)	303,577 (26,514)	112,625 (157,865)
a_1	1.00 (0.01)	316,222 (26,930)	112,635 (157,844)
$2\,a_1$	1.00 (0.01)	343,925 (27,871)	112,757 (158,040)
$3\,a_1$	1.00 (0.01)	373,822 (29,214)	112,910 (158,749)
$4\,a_1$	1.00 (0.01)	406,551 (30,439)	112,767 (159,015)
$5\,a_1$	1.00 (0.01)	441,710 (31,530)	112,592 (158,793)

Table 3: Summary statistics for the expected (i) exit time, (ii) fees collected, and (iii) impermanent loss, as a_1 varies. Mean values (with standard deviation in parenthesis) across 10,000 simulations.

Impact of intensity changes (a_2)

a_2^{new}	$\mathbb{E}[\tau]$	$\mathbb{E}[R_\tau]$	$\mathbb{E}[\text{IL}_\tau]$
$a_2/5$	0.86 (0.21)	118,244 (33,981)	83,734 (127,349)
$a_2/4$	0.93 (0.13)	142,756 (27,971)	95,493 (139,173)
$a_2/3$	0.98 (0.07)	173,770 (24,588)	104,549 (148,242)
$a_2/2$	0.99 (0.03)	216,953 (24,307)	110,222 (153,239)
a_2	1.00 (0.01)	316,222 (26,930)	112,635 (157,844)
$2\,a_2$	1.00 (0.01)	476,493 (31,957)	113,284 (158,588)
$3\,a_2$	1.00 (0.01)	618,447 (36,240)	113,564 (159,533)
$4\,a_2$	1.00 (0.01)	750,774 (40,318)	113,758 (160,513)
$5\,a_2$	1.00 (0.00)	878,099 (43,924)	113,789 (160,830)

Table 4: Summary statistics for the expected (i) exit time, (ii) fees collected, and (iii) impermanent loss, as a_2 varies. Mean values (with standard deviation in parenthesis) across 10,000 simulations.

Conclusion

Key Takeaway

Key Takeaway

- LPs remain in the pool when fee income is greater than potential impermanent loss, particularly when volatility is low, fees are high, or trading activity is high.

Key Takeaway

- LPs remain in the pool when fee income is greater than potential impermanent loss, particularly when volatility is low, fees are high, or trading activity is high.
- Large price dislocations cause LPs to exit before arbitrageurs realign prices, to avoid incurring large impermanent loss.

Key Takeaway

- LPs remain in the pool when fee income is greater than potential impermanent loss, particularly when volatility is low, fees are high, or trading activity is high.
- Large price dislocations cause LPs to exit before arbitrageurs realign prices, to avoid incurring large impermanent loss.

Future research

Key Takeaway

- LPs remain in the pool when fee income is greater than potential impermanent loss, particularly when volatility is low, fees are high, or trading activity is high.
- Large price dislocations cause LPs to exit before arbitrageurs realign prices, to avoid incurring large impermanent loss.

Future research

- Investigate how transaction costs and priority gas fees affect the competition between LPs and arbitrageurs.

Key Takeaway

- LPs remain in the pool when fee income is greater than potential impermanent loss, particularly when volatility is low, fees are high, or trading activity is high.
- Large price dislocations cause LPs to exit before arbitrageurs realign prices, to avoid incurring large impermanent loss.

Future research

- Investigate how transaction costs and priority gas fees affect the competition between LPs and arbitrageurs.
- Extend this framework to concentrated liquidity pools (e.g.; Uniswap v3)

References

Angeris, G., Kao, H.-T., Chiang, R., Noyes, C., and Chitra, T. (2020). An analysis of uniswap markets. *Cryptoeconomic Systems Journal*.

Aoyagi, J. and Ito, Y. (2025). Coexisting exchange platforms: Limit order books and automated market makers. *Journal of Political Economy Microeconomics*, 3(3):000–000.

Aqsha, A., Bergault, P., and Sánchez-Betancourt, L. (2025). Equilibrium reward for liquidity providers in automated market makers. *arXiv preprint arXiv:2503.22502*.

Baggiani, L., Herdegen, M., and Sánchez-Betancourt, L. (2025). Optimal dynamic fees in automated market makers. *arXiv preprint arXiv:2506.02869*.

Barbon, A. and Ranaldo, A. (2021). On the quality of cryptocurrency markets: Centralized versus decentralized exchanges. *arXiv preprint arXiv:2112.07386*.

Bergault, P., Bertucci, L., Bouba, D., and Guéant, O. (2024). Automated market makers: Mean-variance analysis of lps payoffs and design of pricing functions. *Digital Finance*, 6(2):225–247.

Campbell, S., Bergault, P., Milionis, J., and Nutz, M. (2025). Optimal fees for liquidity provision in automated market makers. *arXiv preprint arXiv:2508.08152*.

Canidio, A. and Fritsch, R. (2023). Arbitrageurs' profits, lvr, and sandwich attacks: batch trading as an amm design response. *arXiv preprint arXiv:2307.02074*.

Cao, D., Falk, B. H., Kogan, L., and Tsoukalas, G. (2025). A structural model of automated market making. *Available at SSRN 4591447*.

Capponi, A. and Zhu, B. (2025). Optimal exiting for liquidity provision in constant function market makers. *Available at SSRN 5148585*.

Cartea, A., Drissi, F., and Monga, M. (2023). Decentralised finance and automated market making: Predictable loss and optimal liquidity provision. (2309.08431).

Cartea, Á., Drissi, F., Sánchez-Betancourt, L., Siska, D., and Szpruch, L. (2024). Strategic bonding curves in automated market makers. *Available at SSRN 5018420*.

Evans, A., Angeris, G., and Chitra, T. (2021). Optimal fees for geometric mean market makers. In *International Conference on Financial Cryptography and Data Security*, pages 65–79. Springer.

Fukasawa, M., Maire, B., and Wunsch, M. (2024). Model-free hedging of impermanent loss in geometric mean market makers with proportional transaction fees. *Applied Mathematical Finance*, 31(2):108–129.

Lehar, A. and Parlour, C. (2025). Decentralized exchange: The uniswap automated market maker. *The Journal of Finance*, 80(1):321–374.

Milionis, J., Moallemi, C. C., Roughgarden, T., and Zhang, A. L. (2022). Automated market making and loss-versus-rebalancing. (2208.06046).